



BIOLOGY CH.21 — SCIENTIFIC STUDY — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Origin of Life & Evolution
2. Branches of Biology & Scientific Studies
3. Medical Techniques & Instruments
4. Taxonomy & Classification
5. Fossils & Living Fossils
6. Chemical Tests
7. Important Scientists & Their Contributions
8. ⚠ Traps / Key Differences

1. Origin of Life & Evolution

- J.B.S. Haldane (1929) ne suggest kiya ki life simple inorganic molecules se early Earth par develop hui. Unhone speculate kiya ki early Earth conditions ne complex organic molecules ko janam diya jo life ke liye necessary hain. [Q1164]
- Alexander Oparin (Russian biochemist) ne chemical evolution theory propose ki - origin of life = simplest primordial life ka nonliving matter se appear hona. [Q1183]
- Charles Darwin ne theory of evolution by natural selection propose ki - book 'On the Origin of Species' (1859). Evolution ko 'descent with modification' define kiya. Alfred Russel Wallace ne bhi saath mein propose kiya tha. [Q1178, Q1182]
- Human evolution Africa se trace hoti hai. Paleoanthropology = human evolution ka scientific study. Stages: Dryopithecus → Ramapithecus (Shivalik) → Australopithecus (South Africa, 1924) → Homo Erectus (Java, 1891) → Homo Sapiens Neanderthalensis → Homo Sapiens. [Q1184]

2. Branches of Biology & Scientific Studies

- Ethology = biological study of animal behaviour. [Q1167]
- Toxicology = study of harmful effects of chemicals/substances/situations on people, animals, environment. Virology = viruses. Pathology = disease. Epidemiology = spread/control of diseases. Bacteriology = bacteria. [Q1172]
- Arachnology = study of spiders, scorpions, pseudoscorpions, harvestmen. Agronomy = soil/water/crop management. Zoology = animals. Anthology = flowers/flowering plants. [Q1173]
- Agrostology = scientific study of grass. Anthropology = humans. Pomology = fruits. Osteology = bones. Ornithology = birds. Odontology = teeth/gums. [Q1175]
- Paleontology = study of fossils (remains of plants, animals, fungi, bacteria). Paleobotany = fossil plants, algae, fungi. Paleobiology = ancient organisms. [Q1176]
- Eugenics = branch aiming to improve genetic quality of human population. Biotechnology = biological systems to develop products. Genetics = genes/heredity. Epigenetics = changes in gene function/expression. [Q1169]

- Molecular phylogeny = hereditary molecular differences in DNA sequences/genetics se evolutionary relationships determine karti hai. Metamorphosis = extreme rapid physical changes after birth. [Q1168]

3. Medical Techniques & Instruments

- Echocardiography = ultrasound waves ke reflection se heart ki images produce karti hai. Ultrasonography = high-frequency sound waves se internal body structures ka diagnostic imaging. Sonography = ultrasound waves se medical imaging. [Q1170]
- Autopsy (Necropsy) = post-mortem body ki internal/external medical examination to determine cause of death. Forensic autopsy = suspicious/violent/unknown causes ke liye. Clinical autopsy = hospital mein pathologist dwara. [Q1171]
- Ultrasound = ESWL (Extracorporeal Shock Wave Lithotripsy) mein use hota hai kidney stones ko shatter karne ke liye. Infrasound = frequencies below 20 Hz. Electromagnetic waves = time-varying electric/magnetic fields, medium nahi chahiye. [Q1174]

4. Taxonomy & Classification

- Carolus Linnaeus (Carl Linnaeus) = Father of Taxonomy. Unhone binomial nomenclature introduce kiya (genus + species, e.g., *Panthera tigris*). Definite hierarchy establish ki: Kingdom → Phylum → Class → Order → Family → Genus → Species. [Q1177, Q1179, Q1181]
- Thomas Cavalier-Smith ne six-kingdom classification propose kiya: Animalia, Protozoa, Fungi, Plantae (including red and green algae), Chromista, Bacteria. [Q1180]
- R.H. Whittaker ne five-kingdom classification di: Monera, Protista, Fungi, Plantae, Animalia. [Q1179]

5. Fossils & Living Fossils

- Ginkgo biloba = one of the oldest living tree species, herbal supplement worldwide. Funaria = Bryophyta (moss). Dryopteris = Dryopteridaceae family (male fern). Silver oak (*Grevillea robusta*) = Proteaceae, eastern coastal Australia. [Q1166]

6. Chemical Tests

- Starch + Iodine solution → blue-black color. Rice flour mein starch hota hai, isliye iodine daalne par blue-black ho jata hai. Amylose in starch beta coils structure banata hai jo iodine ke saath interact karke blue-black color deta hai. [Q1165]

7. Important Scientists & Their Contributions

- J.B.S. Haldane = Origin of life from inorganic substances (1929). [Q1164]
- Charles Darwin = Theory of evolution by natural selection (1859). [Q1178, Q1182]

- Alexander Oparin = Chemical evolution theory. [Q1183]
- Gregor Mendel = Laws of inheritance. [Q1182]
- Carolus Linnaeus = Father of Taxonomy, binomial nomenclature. [Q1177, Q1179, Q1181]
- Thomas Cavalier-Smith = Six-kingdom classification. [Q1180]
- John Dalton = Atomic Theory. [Q1182]
- Johann Dobereiner = Law of Triads. [Q1182]
- Democritus = Coined word 'atom'. [Q1181]
- Aristotle = Father of Zoology. [Q1181]
- Herodotus = Father of History. [Q1181]
- Socrates = Father of Philosophy. [Q1181]

8. ⚠ Traps / Key Differences

- Darwin = Evolution, Mendel = Inheritance. [Q1178, Q1182]
- Ethology = animal behaviour, Etiology = cause of disease, Ethnology = human culture, Entomology = insects. [Q1167]
- Echocardiography = heart-specific ultrasound. Ultrasonography = general internal imaging. [Q1170]
- Autopsy = post-mortem examination. [Q1171]
- Paleontology = fossils. Paleobotany = fossil plants. [Q1176]
- Six kingdoms = Cavalier-Smith. Five kingdoms = Whittaker. [Q1180, Q1179]
- Human evolution started in Africa. [Q1184]
- Starch + Iodine = blue-black. [Q1165]